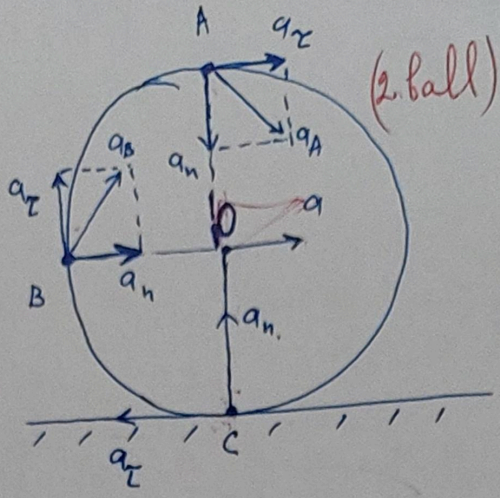
① a

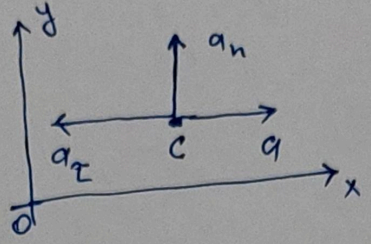
$a_A = ?$

$a_B = ?$



(2 ball) chizmadagi vektorlar to'liq qo'yilgan bo'lsa.

C nuqtada



$a_\tau = a$ ga teng (1 ball)

barcha nuqtalarda.

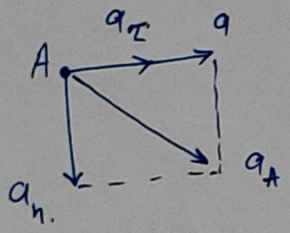
demak OX: $a - a_\tau = 0$ bo'ladi

OY da: a_n qoladi.

demak $a_n = a$ ga teng chiqadi

barcha nuqtalarda. (3 ball)

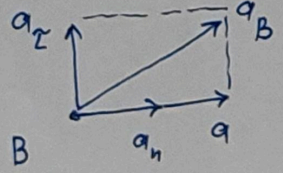
a) A nuqtada.



$a_\tau + a = 2a$

$a_A = \sqrt{a^2 + (2a)^2} = \underline{\sqrt{5} \cdot a}$ (2 ball)

b) B nuqtada.



$a_n + a = 2a$

$a_B = \sqrt{a^2 + (2a)^2} = \underline{\sqrt{5} \cdot a}$ (2 ball)

(2)

(a) $\alpha; v_0; t; g$

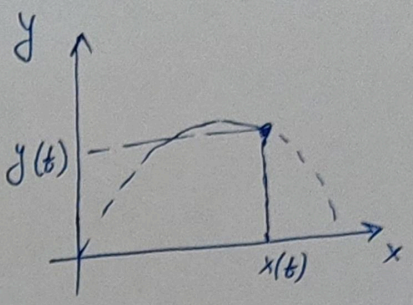
$x(t) = ?$

$y(t) = ?$

$y(x) = ?$

$x(t) = v_0 \cdot \cos\alpha \cdot t$ (0,3)

$y(t) = v_0 \cdot \sin\alpha \cdot t - \frac{gt^2}{2}$ (0,3)



$t = \frac{x}{v_0 \cdot \cos\alpha}$

$y(x) = x \cdot tg\alpha - \frac{gt^2}{2} = x \cdot tg\alpha - \frac{g}{2} \cdot \frac{x^2}{v_0^2 \cos^2\alpha}$ (0,4)

6

(b) $v_0; \alpha; g$

$H_{max} = ?$

$L = ?$

$y'(t) = 0$ $\quad v_0 \cdot \sin\alpha - \frac{g \cdot 2t}{2} = 0.$

$v_0 \cdot \sin\alpha = gt \quad t_k = \frac{v_0 \cdot \sin\alpha}{g}$

$H_{max} = v_0 \cdot \sin\alpha \cdot \frac{v_0 \cdot \sin\alpha}{g} - \frac{g}{2} \cdot \frac{v_0^2 \sin^2\alpha}{g^2} = \frac{v_0^2 \sin^2\alpha}{2g}$ (0,5)

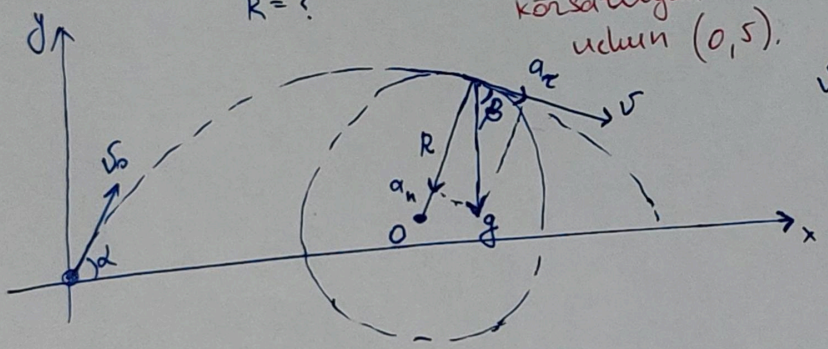
$t_{uch} = 2 \cdot t_k.$

$x(t)$ dan $L = v_0 \cdot \cos\alpha \cdot t_{uch} = v_0 \cos\alpha \cdot 2 \cdot \frac{v_0 \cdot \sin\alpha}{g} = \frac{v_0^2 \cdot \sin(2\alpha)}{g}$ (0,5)

Keltirib chiqarmasdan, to'g'ridan - to'g'ri yozilgan formula uchun ball berilmaydi.

(c) t, α, v_0 | $a_n = ?$ $a_\tau = ?$ $R = ?$

Vektor kattaliklar to'liq ko'rsatilgan chizma uchun (0,5).



$v_x = v_0 \cdot \cos\alpha$

$v_y = v_0 \cdot \sin\alpha - g \cdot t$

$v = \sqrt{v_x^2 + v_y^2} = \sqrt{(v_0\cos\alpha)^2 + (v_0\sin\alpha - gt)^2}$

(1) $a_\tau = \frac{g \cdot (v_0 \sin\alpha - g \cdot t)}{\sqrt{(v_0 \cos\alpha)^2 + (v_0 \sin\alpha - gt)^2}}$ (0,5)

$\cos\beta = \frac{v_y}{v}$ $\quad \cos\beta = \frac{a_\tau}{g}$

$\frac{v_y}{v} = \frac{a_\tau}{g}$ $\quad a_\tau = \frac{g \cdot v_y}{v}$ (1)

(2) $a_n = \frac{g \cdot v_0 \cdot \cos\alpha}{\sqrt{(v_0 \cos\alpha)^2 + (v_0 \sin\alpha - gt)^2}}$ (0,5)

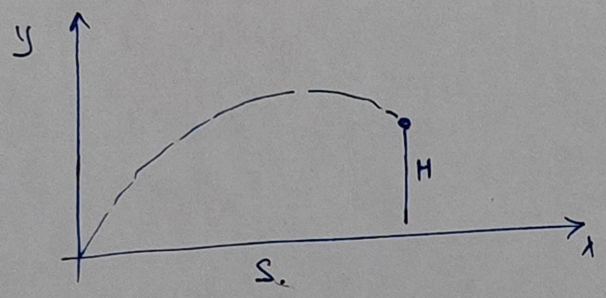
$\sin\beta = \frac{v_x}{v}$ $\quad \sin\beta = \frac{a_n}{g}$

$\frac{v_x}{v} = \frac{a_n}{g}$ $\quad a_n = \frac{g \cdot v_x}{v}$ (2)

$$R = \frac{v^2}{a_n} = \frac{v^2}{g \cdot \frac{v_x}{v}} = \frac{v^3}{g \cdot v_x} \quad (3).$$

$$R = \frac{\left((v_0 \cos\alpha)^2 + (v_0 \sin\alpha - gt)^2\right)^{3/2}}{v_0 \cdot \cos\alpha \cdot g} \quad (0,5).$$

d)



$$H = v_0 \sin\alpha \cdot t - \frac{gt^2}{2}$$

$$S = v_0 \cdot \cos\alpha \cdot t$$

$$\left(H + \frac{gt^2}{2}\right)^2 + S^2 = v_0^2 \cdot t^2$$

$$v_0^2 = \frac{\left(H + \frac{gt^2}{2}\right)^2 + S^2}{t^2}$$ shu yerga cha (0,5)

$$v_0^2 = Hg + \frac{g^2t^2}{4} + \frac{H^2+S^2}{t^2}$$

$v_0' = 0$. Bunga (0,5)

$$\frac{g^2 t}{2} - 2 \cdot \frac{H^2+S^2}{t^3} = 0$$

$$t = \sqrt[4]{\frac{4}{g^2}(S^2+H^2)}$$

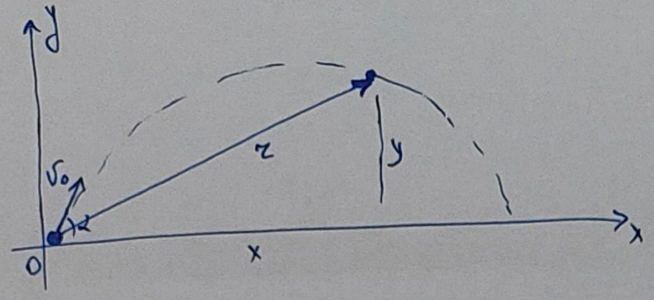
$$v_0^2 = Hg + \frac{g^2 \cdot \sqrt{\frac{4}{g^2}(S^2+H^2)}}{4} + \frac{H^2+S^2}{\sqrt{\frac{4}{g^2}(S^2+H^2)}}$$

$$v_0^2 = Hg + g\sqrt{S^2+H^2}$$

$$v_0 = \sqrt{g\left(H + \sqrt{S^2+H^2}\right)}$$

$$v_{min} = \sqrt{g\left(H + \sqrt{S^2+H^2}\right)}$$ (2) ja'mi

e)



$z^2 = x^2 + y^2$ $\quad x = v_0 \cdot \cos\alpha \cdot t$

$y = v_0 \cdot \sin\alpha \cdot t - \frac{g \cdot t^2}{2}$

z har doim kattalashib borishi uchun

z(t) funksiya o'suvchi bo'lishi kerak.

(0,5) $z'(t) \geq 0$. bo'ladi.

$$z(t) = \sqrt{(v_0 \cos\alpha t)^2 + \left(v_0 \sin\alpha t - \frac{gt^2}{2}\right)^2}$$

shu yergacha (0,5)

$z'(t)$ hosiladan quyidagi natija kelib chiqadi.

Hosila to'g'ri olinib, yechilsa (0,5)

$$2v_0^2 t - 3v_0 \sin\alpha \cdot g \cdot t^2 + g^2 t^3 \geq 0.$$

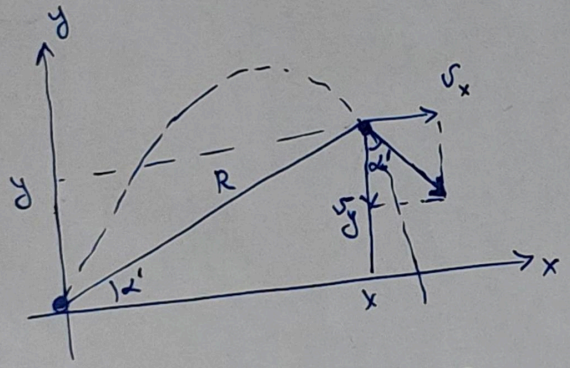
$$3v_0 \sin\alpha\, g t^2 \leq 2v_0^2 t + g^2 t^3$$

$$\sin\alpha \leq \frac{2v_0}{3gt} + \frac{gt}{3v_0} \geq 2\sqrt{\frac{2 \cdot \cancel{v_0}}{3 \cdot \cancel{gt}} \cdot \frac{\cancel{gt}}{\cancel{3v_0}}} = 2\frac{\sqrt{2}}{3}.$$

$a + b \geq 2\sqrt{a \cdot b}$

$\sin\alpha \leq \frac{2\sqrt{2}}{3}$ $\quad \alpha \leq 70,5^\circ$ (4) javob

2-usul:



$tg\alpha' = \frac{y}{x}$ $\quad tg\alpha' = \frac{v_x}{-v_y}$

$$\frac{y}{x} = \frac{v_x}{-v_y}$$

$$\frac{v_0 \sin\alpha \cdot \cancel{t} - \frac{gt^2}{2}}{v_0 \cos\alpha \cdot \cancel{t}} = \frac{v_0 \cos\alpha}{gt - v_0 \sin\alpha}$$

$$v_0 \sin\alpha \cdot gt - \frac{g^2 t^2}{2} - v_0^2 \sin^2\alpha + v_0 \sin\alpha \cdot \frac{gt}{2} = v_0^2 \cos^2\alpha$$

$$\frac{3}{2} v_0 \cdot \sin\alpha \cdot g \cdot t - \frac{g^2 t^2}{2} = v_0^2$$

$$g^2 t^2 - 3v_0 \cdot \sin\alpha \cdot g \cdot t + 2v_0^2 = 0.$$

$D = 9v_0^2 \cdot \sin^2\alpha \cdot g^2 - 4 \cdot g^2 \cdot 2v_0^2$

$D \leq 0$ $\quad v_0^2 \cdot g^2 (9\sin^2\alpha - 8) \leq 0$

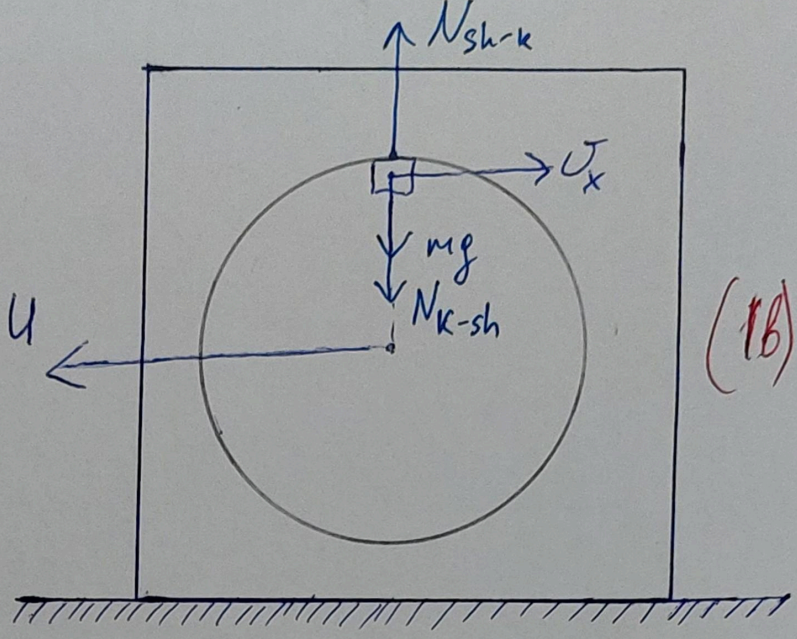
$9\sin^2\alpha \leq 8$

$\sin^2\alpha \leq \frac{8}{9}$.

$\sin_{min} = \frac{2\sqrt{2}}{3}$ $\quad \alpha \leq 70,5^\circ$

3-masala.

$\frac{M}{m} = k$



(1B)

Kubning sirtdan ajrala olishi, shayba-ning chizmada ko'rsatilgan vaziyatga kelishi bilan harakterlash mumkin. ya'ni, dinamik tenglamalardan:

$$\begin{cases} m a_n = mg + N_{k-sh} \\ N_{sh-k} \geq Mg \end{cases}$$

$$a_n = \frac{v_x^2}{R}$$

N_{k-sh} - kubning shaybaga ta'siri

N_{sh-k} - shaybaning kubga ta'siri; aynan shu kuch kubni ko'tarmoqda

Nyutonning III - qonuniga ko'ra $\vec{N}_{sh-k} = -\vec{N}_{k-sh}$

yuqoridagilardan

$$\frac{m v_x^2}{R} \geq mg + Mg \quad (1) \qquad (1B)$$

ekanligi kelib chiqadi:

Impuls saqlanish qonuniga ko'ra

$$mv = Mu - m(v_x - u) \quad (2) \qquad (1,5\ b)$$

Energiya saqlanish qonuniga ko'ra

$$\frac{mv^2}{2} = 2mgR + \frac{Mu^2}{2} + \frac{m(v_x-u)^2}{2} \quad (3) \qquad (1,5\ b)$$

(1), (2) va (3) Tenglamalar sistemasidan g va R ni mavjud va $k=\frac{M}{m}$ ekanligini inobatga olib yechsak,

$$v = \sqrt{gR}\sqrt{5+\frac{4}{k}+k} \qquad (3\ b)$$

kelib chiqadi. Minimum esa:

$$v'(k) = \frac{-\frac{4}{k^2}+1}{\cdots\cdots} = 0 \quad \text{dan}$$

$$k^2 = 4 \rightarrow \boxed{k=2} \qquad (2\ b)$$

4-masala

„defekt" mavjud bōlmagan hol u-n, sferaning q zaryadi uning sirti bōyicha tekis taqsimlanib, sfera markazida

$$\varphi = \frac{q}{4\pi\varepsilon_0 R} \quad (0,5)$$

potensialni hosil ~~qiladi~~ qiladi. Bunda sfera sigimi

$$C = \frac{q}{\varphi} = 4\pi\varepsilon_0 R \quad (0,5)$$

„defekt"ning hisobiga φ potensial $\Delta\varphi$ kichik miqdorga ortadi. Bunda yarimsferada juda kichik q' zaryad <u>joylashgan</u> deb olamiz. (2)

$$\Delta\varphi = \frac{q'}{4\pi\varepsilon_0(R+z)} - \frac{q'}{4\pi\varepsilon_0 R} = -\frac{q'z}{4\pi\varepsilon_0 R(R+z)} \approx$$

$$\approx -\frac{q'z}{4\pi\varepsilon_0 R^2} \quad (1)$$

q' ni topib olish u-n

σ dan foydalanamiz.

$\sigma \simeq const$ (1)

$$\frac{q'}{2\pi z^2} = \frac{q}{4\pi R^2} \rightarrow q' = q \cdot \frac{z^2}{2R^2} \quad (2)$$

$$\Delta C = \frac{q}{\varphi + \Delta\varphi} - \frac{q}{\varphi} = -\frac{q}{\varphi(\varphi+\Delta\varphi)}\Delta\varphi \simeq$$

$$\simeq -\frac{q}{\varphi^2}\cdot\Delta\varphi$$

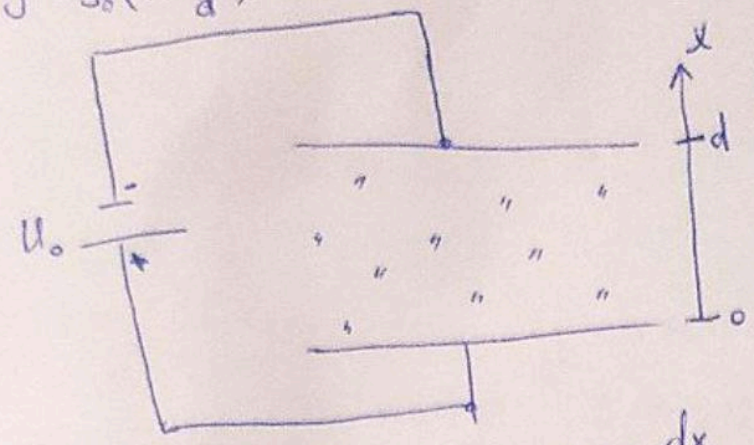
$$\Delta C = -\frac{q}{q^2}(4\pi\varepsilon_0 R)^2 \cdot \left(-\frac{q' z}{4\pi\varepsilon_0 R^2}\right) =$$

$$= \frac{(4\pi\varepsilon_0)^2 R^2}{q} \cdot \frac{q' \cdot z}{4\pi\varepsilon_0 \cdot R^2} = \frac{2\pi\varepsilon_0 z}{q} \cdot q \cdot \frac{z^2}{2R^2}$$

$$\boxed{\Delta C = 2\pi\varepsilon_0 \frac{z^3}{R^2}} \quad (3)$$

U_0; C_0; $\varepsilon = 1$

$\rho = \rho_0\left(1 + \frac{2x}{d}\right)$



① Zanjirda o'zgarmas

$I = const$ tok

paydo bo'ladi.

$dU = I\,dR$

$dR = \rho \cdot \frac{dx}{S}$

$dU = I \cdot \rho \frac{dx}{S}$

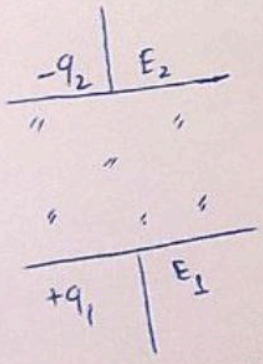
$dU = I \cdot \left(\rho_0 + \rho_0 \cdot \frac{2x}{d}\right) \frac{dx}{S}$ (1)

(1) $U_0 = \int_0^d dU = \int_0^d I \cdot \rho_0 \cdot \left(1 + \frac{2x}{d}\right) \frac{dx}{S} = \int_0^d \frac{I \cdot \rho_0}{S}\left(1 + \frac{2x}{d}\right) dx$

$\int_0^d \left(1 + \frac{2x}{d}\right) = \left(x + \frac{x^2}{d}\right)\Big|_0^d = d + \frac{d^2}{d} = 2d.$

$U_0 = \frac{I \cdot \rho_0}{S} \cdot 2d$ $\qquad C_0 = \frac{\varepsilon_0 \cdot S}{d} \Rightarrow \frac{d}{S} = \frac{\varepsilon_0}{C_0}$

$U_0 = \frac{I \cdot \rho_0 \cdot 2 \cdot \varepsilon_0}{C_0} \Rightarrow \boxed{I = \frac{U_0 \cdot C_0}{2\rho_0 \cdot \varepsilon_0}}$ (1)

② 

Endi kondensatorning yuqori (E_2) va pastki (E_1) tuchlangan liklarini topamiz.

$E(x) = \frac{dU}{dx} = \frac{I \cdot \rho_0}{S}\left(1 + \frac{2x}{d}\right) = \frac{C_0 \cdot U_0}{2\varepsilon_0 S}\left(1 + \frac{2x}{d}\right)$ (1)

$x = 0$ da $E_1 = \frac{C_0 \cdot U_0}{2\varepsilon_0 S}$ $\qquad x = d$ da $E_2 = \frac{3 C_0 U_0}{2\varepsilon_0 S}$

(1) $q_1 = S \cdot \sigma_1 = S \cdot E_1 \cdot \varepsilon_0 = \boxed{\frac{C_0 \cdot U_0}{2}}$

(1) $q_2 = -S \cdot \sigma_2 = -S \cdot E_2 \cdot \varepsilon_0 = \boxed{-\frac{3 C_0 U_0}{2}}$

(3) Kondensatorning umumiy zaryadi: plastinkalardagi (q_1 va q_2), plastinkalar orasidagi (q) zaryadlarning yig'indisiga teng

$$q_1 + q_2 + q = 0 \quad (1)$$

$$q = -q_1 - q_2 = -\frac{C_0 U_0}{2} - \left(-\frac{3 C_0 U_0}{2}\right)$$

$$\boxed{q = C_0 \cdot U_0} \quad (1)$$

(4) Elektr maydon energiyasini esa, kondensator-ning elektr maydon energiya zichligi formulasidan topamiz:

$$\omega = \frac{\varepsilon_0 \cdot E^2}{2} \qquad (1)\ \int dW = \int \omega \cdot dV \qquad dV = S\,dx$$

$$W = \int \omega\, dV = \int_0^d \frac{\varepsilon_0 E^2(x)}{2} \cdot S\,dx = \int_0^d \frac{\varepsilon_0 \left(\frac{C_0 \cdot U_0}{2\varepsilon_0 S}\left(1 + \frac{2x}{d}\right)\right)^2}{2} S\,dx$$

$$W = \frac{C_0 \cdot U_0^2}{8d} \int_0^d \left(1 + \frac{2x}{d}\right)^2 dx = \boxed{\frac{13 \cdot C_0 \cdot U_0^2}{24}} \quad (1)$$